

Harsh Shrivastava

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Research Interests

Foundations of Causal Inference · Causal Representation Learning and Discovery · Robust and Generalizable AI

Education

- Aug 2025 – Present** • **Mohamed bin Zayed University of Artificial Intelligence (MBZUAI)**, UAE
M.Sc. in Machine Learning CGPA: 3.61/4.00
- Aug 2016 – Aug 2020** • **Rustamji Institute of Technology (RJIT)**, India
B.Eng. in Computer Science and Engineering CGPA: 8.36/10

Selected Papers

- UAI 2026** • **Fixed-Confidence Best-Arm Identification for Causal Mediation Analysis**
Harsh Shrivastava, Yuta Kawakami, Junpei Komiyama, and Jin Tian.
UAI 2026. [Proceedings] [arXiv]
- Under Review** • **Evaluating Joint Counterfactual Functionals from Marginal Distributions and Edgewise Correlations**
Naoya Hashimoto, Harsh Shrivastava, Yuta Kawakami, and Jin Tian.
Submitted to AAAI 2027.
- ECCV 2026** • **MoVA: Learning Asymmetric Dual Projections for Modular Long Video-Text Alignment**
Peiyuan Zhu, Shaoan Xie, Zijian Li, Yifan Shen, Namrata Deka, Harsh Shrivastava, Guangyi Chen, and Kun Zhang.
ECCV 2026, to appear.

Research Experience

- Aug 2025 – Present** • **Mohamed bin Zayed University of Artificial Intelligence (MBZUAI)**, UAE
Object-Centric Sparse Transition Representations for World Models.
Advisor: Prof. Kun Zhang.
Developing object-centric latent representations with sparse transition dependencies for structured world models, with the goal of improving the modularity and generalization of learned dynamics.
Fixed-Confidence Best-Arm Identification for Causal Mediation Analysis.
Advisors: Prof. Jin Tian and Prof. Junpei Komiyama; Mentor: Dr. Yuta Kawakami.
Formulated maximization of the expected natural direct potential outcome as a fixed-confidence best-arm identification problem. Established identification from interventional distributions, derived instance-dependent sample-complexity lower bounds, and developed an asymptotically optimal Track-and-Stop algorithm for adaptive treatment allocation.
Evaluating Joint Counterfactual Functionals from Marginal Distributions and Edgewise Correlations.
Advisor: Prof. Jin Tian; Collaborators: Naoya Hashimoto and Dr. Yuta Kawakami.
Studying how joint counterfactual functionals can be evaluated when only marginal distributions and edgewise correlations are available, without access to the full joint counterfactual distribution. Contributing to methods that use this partial dependence information to constrain otherwise unidentified counterfactual quantities.
- Aug 2019 – Mar 2020** • **National University of Singapore (NUS)**, Singapore
Research Assistant; Advisors: Prof. Roger Zimmermann and Dr. Yifang Yin.
Developed label-noise-robust audio representations using multi-task graph convolutional networks and a multimodal teacher-student framework for audio tagging, leading to publications at ICASSP 2020 and AAAI 2021.
- Dec 2018 – Jun 2019** • **Indraprastha Institute of Information Technology Delhi (IIIT-Delhi)**, India
Research Intern; Advisor: Prof. Rajiv Ratn Shah.
Developed a human-in-the-loop face-retrieval system and a multi-level feature-fusion model for weakly supervised audio tagging, leading to publications at IEEE BigMM 2019 and ACM Multimedia 2019.

Additional Publications

- AAAI 2021** • **Enhanced Audio Tagging via Multi- to Single-Modal Teacher-Student Mutual Learning**
Y. Yin, Harsh Shrivastava, Y. Zhang, Z. Liu, R. R. Shah, and R. Zimmermann.
AAAI Conference on Artificial Intelligence (AAAI), 2021.
- ICASSP 2020** • **Multi-task Graph Convolutional Network for Audio Tagging with Noisy Labels**
Harsh Shrivastava, Y. Yin, R. R. Shah, and R. Zimmermann.
IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2020.
- BigMM 2019** • **FaceFetch: An Efficient and Scalable Face Retrieval System That Uses Your Visual Memory**
Harsh Shrivastava, R. PVNS, K. Aggarwal, M. Ayyar, Y. Yin, R. R. Shah, and R. Zimmermann.
IEEE International Conference on Multimedia Big Data (BigMM), 2019.
- ACM MM 2019** • **Multi-Level Fusion Based Class-Aware Attention Model for Weakly Labeled Audio Tagging**
Y. Yin, M.-J. Chiou, W. Zhenguang, Harsh Shrivastava, R. R. Shah, and R. Zimmermann.
ACM International Conference on Multimedia (ACM MM), 2019.

Industry Experience

- Apr 2024 – Jul 2025** • **Ecom Express, India** *Data Scientist II*
Developed sequential prediction models from large-scale shipment trajectories for logistics planning, routing, and delivery-time estimation.
- Jul 2023 – Mar 2024** • **Farfetch, India** *Data Scientist*
Built contextual recommendation and ranking systems for product search and discovery, addressing cold-start settings and learning from logged user interactions.
- Aug 2021 – Jun 2023** • **Flipkart, India** *Data Scientist*
Developed learning-to-rank and multi-objective optimization methods for large-scale autosuggestion systems under partial user feedback. Received the FK Search RnR Award in 2022.
- Sep 2020 – Aug 2021** • **CARS24, India** *Data Scientist*
Developed and deployed pricing and demand-prediction models for an online automotive marketplace.

Technical Skills

- Programming** • Python, SQL, PySpark
- ML and Scientific Computing** • PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn, NumPy, Pandas
- Tools and Cloud** • Git, Docker, GCP, AWS, Azure